Note

In congruency, the coinciding parts are called corresponding parts. The coinciding sides are called corresponding sides and the coinciding angles are called corresponding angles.

So, if the corresponding sides and corresponding angles of two plane figures are equal then they are called congruent figures. If two plane figures F_1 and F_2 are congruent, we can write $F_1 \cong F_2$

4.4.5 Congruence of Triangles

Triangle is a closed figure formed by three line segments. There are three sides and three angles. If the corresponding sides and corresponding angles of two triangles are equal, then the two triangles are said to be congruent to each other.

Observe the following two triangles $\triangle ABC$ and $\triangle XYZ$.

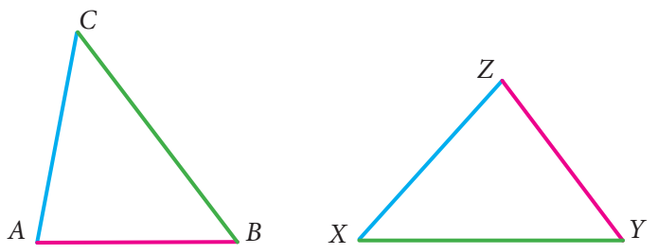


Fig. 4.22

By trace copy superposition method, it can be observed that $\triangle XYZ$ matches identically with $\triangle ABC$.

All the sides and the angles of $\triangle ABC$ are equal to the corresponding sides and angles of $\triangle XYZ$. So we can say the two triangles are congruent. We can express it as $\triangle ABC \cong \triangle XYZ$.

We can also observe that the vertex A matches with vertex Z, B matches with Y and C matches with X. They are called corresponding vertices.

The side AB matches YZ, BC matches XY and CA matches ZX. These pairs are called corresponding sides.

Also, $\angle A = \angle Z$, $\angle B = \angle Y$ and $\angle C = \angle X$. These pairs are called corresponding angles.

In the above case, the correspondance is $A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X$. We write this as $ABC \leftrightarrow XYZ$.

Suppose we try to match vertex A with vertex Y or vertex X (Fig. 4.23), we can observe that the triangles do not match each other.

7th_Maths_T2_EM_Chp4.indd 78 29-05-2019 12:19:39